GPU value appears at larger study sizes

Same four models and five outputs per scenario • JAX on CPU vs RTX 2080 Ti

8,192 scenarios

4.2× faster

GPU vs optimized CPU, warm run

GPU batch time

16.4 ms

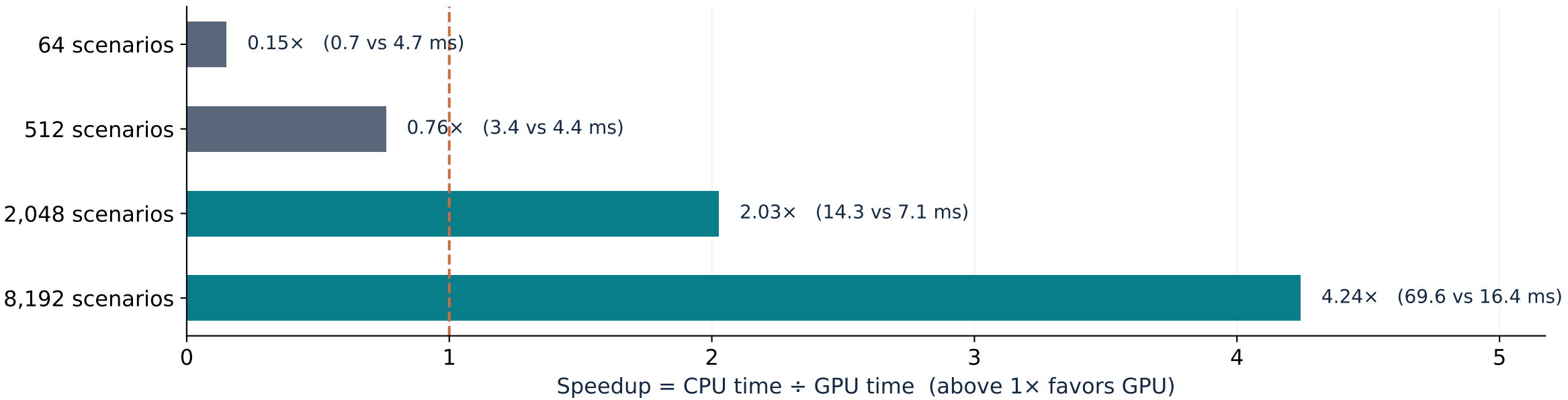
CPU: 69.6 ms; data round trip included

Measured large batches

≥2,048

GPU wins at measured larger batches

Measured speedup by batch size



TIMING SCOPE

Median synchronized runs; input transfer and output copy included. Compilation excluded from bars.

First call took 0.6–0.7 s on CPU and 0.7–1.0 s on GPU across measured sizes.

Machine: Ryzen 5 5600X, RTX 2080 Ti, WSL2; exact timings vary with load and hardware.